

Genesis Humanoid Robot – Arm Design Reference

Team: Genesis

Competition: RoboCup@Home 2027

Subsystem: Arm

Status: Preliminary Calculations and actuators selection for rough estimations

1. Purpose

This document summarizes the current arm architecture, calculations, actuator selection, and CAD-relevant parameters.

It serves as a common reference for the CAD/Mechanical, Electronics, and Controls teams and will be updated as the design progresses.

Workflow:

Requirement → ***Calculation*** → ***Actuator Selection*** → ***CAD*** → ***Prototype*** → ***Testing*** → ***Validation***

2. Arm Configuration

Total: 7 DOF			
Section	Joint	DOF	
Shoulder	Pitch	1	
Shoulder	Roll	1	
Shoulder	Yaw	1	
Elbow	Revolute	1	
Wrist	Pitch	1	
Wrist	Roll	1	
Wrist	Yaw	1	

Current Arm Dimensions

- Upper arm: 31 cm
- Elbow-to-wrist: 28 cm
- Gripper: 16 cm
- Maximum elbow angle: 120°
- Approx. fully extended shoulder-to-gripper length: 75 cm
- Target payload: ~2.5 kg
- Factor of Safety: 1.5

3. Torque, RPM & Actuator Selection

Joint	Req. Torque	Req. RPM	Actuator	Rated Torque	Rated RPM	
Shoulder Pitch	65 N·m	~30	CubeMars AKH70-48	74 N·m	28	
Shoulder Roll	55 N·m	~30	CubeMars AKE90-8	55 N·m	120	
Shoulder Yaw	~14–16 N·m	~30	CubeMars AK60-39 V3.0	24 N·m	70	
Elbow	24 N·m	30	CubeMars AK60-39 V3.0	24 N·m	70	
Wrist Pitch	7.1 N·m	30	DS3245	TBD	TBD	
Wrist Roll	7.1 N·m	30	DS3245	TBD	TBD	
Wrist Yaw	2–3 N·m	30	Dynamixel XM430-W350-R	—	—	
Gripper	3.24 N·m	45	Feetech SM150	4.5 N·m	—	

NOTE: Using DS3245 servo for gripper also

4. CAD Actuator Reference

Actuator	Application	Current CAD Envelope
AKH70-48	Shoulder Pitch	Ø90 × 81.5 mm
AKE90-8	Shoulder Roll	Ø107.5 × 43.5 mm
AK60-39 V3.0	Shoulder Yaw / Elbow	Ø79 × 67 mm
DS3245	Wrist Pitch / Roll / Gripper	TBD
XM430-W350-R	Wrist Yaw	28.5 × 46.5 × 34 mm

CAD Note: Final actuator packaging must also consider mounting holes, shaft/interface, bearings, connectors, cable routing, clearances and serviceability.

5. Joint Summary

Shoulder

3 DOF: Pitch, Roll, Yaw

The shoulder is the highest-torque section of the arm and will strongly influence the overall arm size, weight and structural design.

Shoulder Pitch

- Required Torque: **65 N·m**
- Required Speed: **~30 RPM**
- Actuator: **CubeMars AKH70-48**
- Continuous Torque: **74 N·m**
- Rated RPM: **28 RPM**
- Peak Torque: **222 N·m**
- Weight: **1.396 kg**
- Dimensions: **Ø90 × 81.5 mm**

Shoulder Roll

- Required Torque: **55 N·m**
- Required Speed: **~30 RPM**
- Actuator: **CubeMars AKE90-8**
- Continuous Torque: **55 N·m**
- Rated RPM: **120 RPM**
- Peak Torque: **170 N·m**
- Weight: **1.40 kg**
- Dimensions: **Ø107.5 × 43.5 mm**

Shoulder Yaw

- Required Torque: **14–16 N·m**
- Required Speed: **~30 RPM**
- Actuator: **CubeMars AK60-39 V3.0**
- Continuous Torque: **24 N·m**
- Rated RPM: **70 RPM**
- Peak Torque: **72 N·m**
- Weight: **0.75 kg**
- Dimensions: **Ø79 × 67 mm**

Elbow

1 DOF Revolute

- Maximum angle: 120°
- Required torque: 24 N·m
- Required speed: 30 RPM
- Actuator: AK60-39 V3.0

Wrist

3 DOF: Pitch, Roll, Yaw

- Pitch → 7.1 N·m, 30 RPM → DS3245
- Roll → 7.1 N·m, 30 RPM → DS3245
- Yaw → 2–3 N·m, 30 RPM → XM430-W350-R

Keep wrist mass as low as possible to reduce elbow and shoulder torque requirements.

Gripper

- Required torque: 3.24 N·m
- Required speed: 45 RPM
- Actuator: Feetech SM150

6. Cosero – Reference Robot

Cosero is being used as a reference because it was developed for domestic service robotics and RoboCup@Home.

Arm

- 7-DOF anthropomorphic arm
- Two arms
- Reported payload: ~1.5 kg/arm

Reported ActuatorsGripper

Actuator	Holding Torque	Weight
Dynamixel EX-106+	10.7 N·m	154 g
Dynamixel RX-64	6.4 N·m	116 g

Uses an adaptive FinGripper / Fin Ray based concept for conforming to different object shapes.

Relevant Design Lessons

- 7-DOF anthropomorphic configuration
- Lightweight arm construction
- Compact actuator integration
- ~1.5 kg/arm payload benchmark
- Adaptive gripper concept

Cosero is a reference, not a direct copy

8. Current Status

Item	Status
Arm architecture	✅ 7 DOF
Elbow	✅ 1 DOF Revolute
Arm dimensions	✅ Preliminary
FOS	✅ 1.5
Torque calculations	✅ Preliminary
RPM calculations	✅ Preliminary
Actuator selection	🟡 Preliminary
Wrist actuator	🟡 DS3245 selected
CAD	🟡 In progress
Bearings	⌚ Pending
Dynamic validation	⌚ Pending
Prototype testing	⌚ Pending

9. CAD Quick Reference

7 DOF | ~2.5 kg payload | FOS 1.5

Upper arm: 31 cm

Elbow-to-wrist: 28 cm

Gripper: 16 cm

Max elbow angle: 120°

Fully extended length: ~75 cm

Actuators:

Shoulder Pitch → AKH70-48

Shoulder Roll → AKE90-8

Shoulder Yaw → AK60-39 V3.0

Elbow → AK60-39 V3.0

Wrist Pitch → DS3245

Wrist Roll → DS3245

Wrist Yaw → XM430-W350-R

Gripper → SM150